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Inventor(s)	Hamill; Whittaker Ian et al.

Systems and methods for protecting the cerebral vasculature

Abstract

Disclosed are methods and devices for isolating all three of the left subclavian, left common carotid and brachiocephalic arteries from embolic debris that might flow through the aortic arch, via a single access point. A system may include an elongate flexible tubular sheath, having a proximal end and a distal end, and an inner member extending through the sheath and moveable relative to the sheath. A left subclavian element may be supported by the inner member. A filter membrane may be configured to isolate the aorta from the brachiocephalic, left common carotid and left subclavian arteries when the left subclavian element is expanded within the left subclavian artery and the sheath is retracted to expose the membrane. The left subclavian element may include a self expandable frame, which may carry a left subclavian filter.

Inventors: **Hamill; Whittaker Ian (Petaluma, CA), Yuan; Shih Hsiung Albert (Pleasanton, CA), Fields; Antony J. (San Francisco, CA), Purcell; Cameron Paul (Santa Rosa, CA), Fifer; Daniel Wayne (Windsor, CA), Watson; James Richard (Santa Rosa, CA)**

Applicant: **Claret Medical, Inc.** (Santa Rosa, CA)

Family ID: **66867757**

Assignee: **Claret Medical, Inc. (Santa Rosa, CA)**

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Primary Examiner: Shi; Katherine M

Assistant Examiner: Adam; Mohammed S

Attorney, Agent or Firm: Seager, Tufte & Wickhem LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 16/395,412, filed Apr. 26, 2019, which claims the benefit of priority under 35 U.S.C. § 119 to U.S. Provisional Application Ser. No. 62/663,117, filed Apr. 26, 2018, the entirety of which is incorporated herein by reference.

TECHNICAL FIELD
(1) In general, the present disclosure relates to medical devices for filtering blood. And, more particularly, in certain embodiments, to a method and a system of filters and deflectors for protecting the cerebral arteries from emboli, debris and the like dislodged during an endovascular or cardiac procedure.
BACKGROUND
(2) Thromboembolic disorders, such as stroke, pulmonary embolism, peripheral thrombosis, atherosclerosis, and the like affect many people. These disorders are a major cause of morbidity and mortality in the United States and throughout the world. Thromboembolic events are characterized by an occlusion of a blood vessel. The occlusion can be caused by a clot which is viscoelastic (jelly-like) and comprises platelets, fibrinogen, and other clotting proteins.
(3) Percutaneous aortic valve replacement procedures have become popular, but stroke rates related to this procedure are between four and twenty percent. During catheter delivery and valve implantation, plaque or other material may be dislodged from the vasculature and may travel

through the carotid circulation and into the brain. When an artery is occluded by a clot or other embolic material, tissue ischemia (lack of oxygen and nutrients) develops. The ischemia progresses to tissue infarction (cell death) if the occlusion persists. Infarction does not develop or is greatly limited if the flow of blood is reestablished rapidly. Failure to reestablish blood-flow can lead to the loss of limb, angina pectoris, myocardial infarction, stroke, or even death.
(4) Various surgical or endovascular techniques and medicaments to remove or dissolve obstructing material have been developed, to reestablish blood flow. However, additional procedures may be traumatic and are best avoided when possible. Additionally, the use of certain devices carry risks such as the risk of dislodging foreign bodies, damaging the interior lining of the vessel as the catheter is being manipulated, blood thinning, etc.
(5) A variety of filtration or deflection devices have been proposed, to prevent entry of debris into the cerebral circulation. Some isolate only the brachiocephalic artery and left common carotid, while others might additionally isolate the left Subclavian but typically through the use of multiple catheters. Others are said to isolate all three arteries leading to the cerebral circulation, from a single catheter, but the catheter is introduced via the femoral artery and none have achieved adoption.
(6) The need thus remains for a simple, single catheter to enable endovascular isolation of the complete cerebral circulation, preferably from an access point other than the femoral artery.
SUMMARY
(7) The present invention provides a three vessel cerebral protection catheter, for introduction through the brachiocephalic artery via the right radial or right brachial artery, and across the aortic arch. A distal element of the catheter is anchored in the left subclavian artery, and a membrane is deployed such that the single catheter isolates all three of the left subclavian, left common carotid and brachiocephalic arteries from embolic debris that might flow through the aortic arch.
(8) In one implementation, a catheter may comprise an elongate flexible tubular sheath, having a proximal end and a distal end, and an inner member extending through the sheath and moveable relative to the sheath, and supporting a left subclavian element. A filter membrane may be carried between the inner member and the sheath, the membrane and the left subclavian element configured to isolate the aorta from the brachiocephalic, left common carotid and left subclavian arteries when the left subclavian element is expanded within the left subclavian artery and the sheath is retracted to expose the membrane. An aortic support hoop may be provided, configured to seal the membrane against the wall of the aorta.
(9) The left subclavian element may comprise a self-expanding frame and may carry a filter membrane which may have a conical configuration. The catheter may further comprise at least one pull wire for laterally deflecting a distal steering zone on the sheath.
(10) There is also provided a method of isolating the cerebral circulation from embolic debris. The method may comprise the steps of advancing a catheter through the brachiocephalic artery and into the left subclavian artery; deploying an anchor within the left subclavian artery; and retracting an outer sheath on the catheter to expose a filter extending from the anchor, across the ostium to the left common carotid artery and into the brachiocephalic artery. The filter may comprise a body, a peripheral edge and a self-expandable frame carried by at least a portion of the peripheral edge; wherein the frame brings the edge into proximity of the wall of the aorta in response to the retracting step. The frame may be in the form of a loop.
(11) The step of advancing a catheter may include the step of proximally retracting a pull wire to laterally deflect a distal portion of the catheter to enter the ostium of the left subclavian artery. The anchor may be carried by a tubular inner member, and the retracting step may comprise retracting the outer sheath relative to the inner member to expose the anchor. The deploying an anchor step may comprise deploying a self-expandable frame, and the self-expandable frame may carry a left subclavian filter.
(12) In a first example, an embolic protection system for isolating the cerebral vasculature may

comprise an elongate outer sheath, having a proximal end and a distal end, an inner member extending through a lumen of the outer sheath, a distal anchoring mechanism coupled to a distal end region of the inner member, a proximal filter membrane carried between the inner member and the outer sheath, the proximal filter membrane configured to extend from the left subclavian artery to the brachiocephalic artery, an aortic support hoop coupled to the proximal filter membrane and configured to seal the proximal filter membrane against the wall of the aorta, and a handle including a first actuation mechanism coupled to proximal end of the inner member and a second actuation mechanism coupled to the proximal end of the outer sheath.

(13) Alternatively or additionally to any of the examples above, in another example, the distal anchoring mechanism comprise a self-expanding frame.

(14) Alternatively or additionally to any of the examples above, in another example, the embolic protection system may further comprise a distal filter membrane coupled to the self-expanding frame.

(15) Alternatively or additionally to any of the examples above, in another example, the embolic protection system may further comprise a proximal bond tube coupled to a proximal filter bag termination of the proximal filter membrane.

(16) Alternatively or additionally to any of the examples above, in another example, the embolic protection system may further comprise an aortic ring connector extending from a proximal end of the aortic ring to proximal bond tube.

(17) Alternatively or additionally to any of the examples above, in another example, the embolic protection system may further comprise a delivery wire coupled to the tube and extending proximally to the handle.

(18) Alternatively or additionally to any of the examples above, in another example, the handle may further comprise a third actuation mechanism coupled to a proximal end of the delivery wire.

(19) Alternatively or additionally to any of the examples above, in another example, the inner member may be slidably disposed within a lumen of the proximal bond tube.

(20) Alternatively or additionally to any of the examples above, in another example, the embolic protection system may further comprise a distal support strut extending between a distal end of the aortic ring and the self-expanding frame of the distal anchoring mechanism.

(21) Alternatively or additionally to any of the examples above, in another example, the embolic protection system may further comprise an aortic ring connector extending from a proximal end of the aortic ring to a connection tube disposed about the inner member and the aortic ring connector, the connection tube distal to a proximal filter bag termination of the proximal filter membrane.

(22) Alternatively or additionally to any of the examples above, in another example, the embolic protection system may further comprise a proximal bond tube coupled to the proximal bag termination of the proximal filter membrane and a delivery wire coupled to the proximal bond tube and extending proximally to the handle.

(23) Alternatively or additionally to any of the examples above, in another example, the embolic protection system may further comprise an aortic ring deployment wire coupled to the aortic ring connector and extending proximally to the handle.

(24) Alternatively or additionally to any of the examples above, in another example, the outer sheath, the inner member, the delivery wire, and the aortic ring deployment wire may be individually actuatable.

(25) Alternatively or additionally to any of the examples above, in another example, a distal end region of the outer sheath may be deflectable.

(26) Alternatively or additionally to any of the examples above, in another example, a cross-sectional shape of the inner member adjacent to the proximal bond tube may be non-circular and at least a portion of the proximal bond tube may have a non-circular cross-sectional shape.

(27) In another illustrative example, an embolic protection system for isolating the cerebral vasculature may comprise an elongate outer sheath, having a proximal end and a distal end, an

inner member extending through a lumen of the outer sheath, a distal filter assembly coupled to a distal end region of the inner member, the distal filter assembly including a self-expanding frame and a distal filter membrane coupled to the self-expanding frame, a proximal filter membrane carried between the inner member and the outer sheath, the proximal filter membrane configured to extend from the left subclavian artery to the brachiocephalic artery, an aortic support hoop coupled to the proximal filter membrane and configured to seal the proximal filter membrane against the wall of the aorta, a proximal bond tube coupled to a proximal filter bag termination of the proximal filter membrane, an aortic ring connector coupled to and extending between a proximal end of the aortic ring and the proximal bond tube, a delivery wire coupled to and extending proximally from the proximal bond tube, and a handle including a first actuation mechanism coupled to proximal end of the inner member, a second actuation mechanism coupled to the proximal end of the outer sheath, and a third actuation mechanism coupled to a proximal end of the delivery wire.
(28) Alternatively or additionally to any of the examples above, in another example, the embolic protection system may further comprise a distal anchoring structure positioned distal to the distal filter assembly.
(29) Alternatively or additionally to any of the examples above, in another example, the aortic ring connector may include a curved shape.
(30) In another example, a method of isolating cerebral circulation from embolic debris may comprise advancing an embolic protection system through the brachiocephalic artery and into the left subclavian artery, deploying a distal anchor within the left subclavian artery, and retracting an outer sheath of the embolic protection system to expose a filter extending from the distal anchor, across the ostium to the left common carotid artery and into the brachiocephalic artery.
(31) Alternatively or additionally to any of the examples above, in another example, the distal anchor may be carried by an inner member, the inner member separately actuatable from the outer sheath.
(32) The above summary of exemplary embodiments is not intended to describe each disclosed embodiment or every implementation of the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The invention may be more completely understood in consideration of the following detailed description of various embodiments in connection with the accompanying drawings, in which:
(2) FIGS. **1-4** illustrate a method deploying a filter system to protect the cerebral vascular architecture.
(3) FIG. **5**A illustrates a first embodiment of a filter system.
(4) FIG. **5**B illustrates another alternative embodiment a filter system.
(5) FIG. **6** is a partial cross-sectional view of the illustrative filter system of FIG. **5**A, taken at line **6-6**.
(6) FIG. **7** is a cross-sectional view of the illustrative filter system of FIG. **5**A, taken at line **7-7**.
(7) FIG. **8** is a cross-sectional view of the illustrative filter system of FIG. **5**A, taken at line **8-8**.
(8) FIG. **9** illustrates another alternative embodiment a filter system.
(9) FIG. **10** is a cross-sectional view of the illustrative filter system of FIG. **9**, taken at line **10-10**.
(10) FIG. **11** is a partial cross-sectional view of the illustrative filter system of FIG. **9**, taken at line **11-11**.
(11) FIG. **12** illustrates another alternative embodiment a filter system.
(12) FIG. **13** illustrates another alternative embodiment a filter system.
(13) FIG. **14** is a cross-sectional view of the illustrative filter system of FIG. **13**, taken at line **13-13**.

(14) FIG. **15** is a partial cross-sectional view of the illustrative filter system of FIG. **13**, taken at line **15-15**.
(15) FIG. **16** is a cross-sectional view of an alternative deployment tube;
(16) FIG. **17** is a partial side view of the illustrative deployment tube of FIG. **16**.
(17) FIG. **18** is a cross-sectional view of an alternative inner member keying system.
(18) FIGS. **19-21** illustrate alternative distal anchoring mechanisms.
(19) FIGS. **22**A-**22**C illustrate alternative aortic ring connector configurations.
(20) FIG. **23** illustrates an alternative aortic ring connector configuration.
(21) While the invention is amenable to various modifications and alternative forms, specifics thereof have been shown by way of example in the drawings and will be described in detail. It should be understood, however, that the intention is not to limit aspects of the invention to the particular embodiments described. On the contrary, the intention is to cover all modifications, equivalents, and alternatives falling within the spirit and scope of the invention.

DETAILED DESCRIPTION

(22) For the following defined terms, these definitions shall be applied, unless a different definition is given in the claims or elsewhere in this specification.
(23) All numeric values are herein assumed to be modified by the term “about”, whether or not explicitly indicated. The term “about” generally refers to a range of numbers that one of skill in the art would consider equivalent to the recited value (i.e., having the same function or result). In many instances, the term “about” may be indicative as including numbers that are rounded to the nearest significant figure.
(24) The recitation of numerical ranges by endpoints includes all numbers within that range (e.g., 1 to 5 includes 1, 1.5, 2, 2.75, 3, 3.80, 4, and 5).
(25) Although some suitable dimension ranges and/or values pertaining to various components, features and/or specifications are disclosed, one of skill in the art, incited by the present disclosure, would understand desired dimensions, ranges and/or values may deviate from those expressly disclosed.
(26) As used in this specification and the appended claims, the singular forms “a”, “an”, and “the” include plural referents unless the content clearly dictates otherwise. As used in this specification and the appended claims, the term “or” is generally employed in its sense including “and/or” unless the content clearly dictates otherwise.
(27) The following detailed description should be read with reference to the drawings in which similar elements in different drawings are numbered the same. The detailed description and the drawings, which are not necessarily to scale, depict illustrative embodiments and are not intended to limit the scope of the invention. The illustrative embodiments depicted are intended only as exemplary. Selected features of any illustrative embodiment may be incorporated into an additional embodiment unless clearly stated to the contrary.
(28) The disclosure generally relates to devices and methods for filtering fluids and/or deflecting debris contained within fluids, including body fluids such as blood. The filtering or deflecting device can be positioned in an artery upstream from the brain before and/or during an endovascular procedure (e.g., transcatheter aortic valve implantation (TAVI) or replacement (TAVR), transcatheter mitral valve implantation (TAMI) or replacement (TAMR), surgical aortic valve replacement (SAVR), other surgical valve repair, implantation, or replacement, cardiac ablation (e.g., ablation of the pulmonary vein to treat atrial fibrillation) using a variety of energy modalities (e.g., radio frequency (RF), energy, cryo, microwave, ultrasound), cardiac bypass surgery (e.g., open-heart, percutaneous), transthoracic graft placement around the aortic arch, valvuloplasty, etc.) to inhibit or prevent embolic material such as debris, emboli, thrombi, etc. resulting from entering the cerebral vasculature.
(29) Devices and methods have been developed to filter blood flowing to the innominate artery and the left common carotid artery, which provide about 90% of the blood entering the cerebral

vasculature. Examples are provided in U.S. Pat. No. 8,876,796, which is herein incorporated by reference in its entirety. Certain such devices and methods leave the left subclavian artery, and thus the left vertebral artery, which provides about 10% of the blood entering the cerebral vasculature, exposed to potential embolic material. Other embodiments described in U.S. Pat. No. 8,876,796 filter blood flowing to the left common carotid artery and the left subclavian artery **16**. Certain such devices and methods leave the innominate artery **12**, and thus both the right common carotid artery **18** and the right vertebral artery **20**, which provide even about 50% of the blood entering the cerebral vasculature, exposed to potential embolic material. Assuming perfect use and operation, either of these options may leave potential stroke rates as high as two to ten percent due to exposed arteries that provide blood flow to the cerebral vasculature.

(30) Several single-access multi-vessel embodiments of cerebral protection devices that can provide full cerebral protection (e.g., protecting all four blood vessels supplying blood to the brain) with minimal arch interference are described below. The devices may be used to trap and/or deflect particles in other blood vessels within a subject, and they can also be used outside of the vasculature. The devices described herein are generally adapted to be delivered percutaneously to a target location within a subject but can be delivered in any suitable way and need not be limited to minimally-invasive procedures.

(31) FIG. **1** is a schematic view of an aortic arch **10**. The aortic arch **10** is downstream of the aortic valve **11**. The aortic arch **10** typically includes three great branch arteries: the brachiocephalic artery or innominate artery **12**, the left common carotid artery **14**, and the left subclavian artery **16**. The innominate artery **12** branches to the right carotid artery **18**, then the right vertebral artery **20**, and thereafter is the right subclavian artery **22**. The right subclavian artery **22** supplies blood to and may be directly accessed from (termed right radial access) the right arm. The left subclavian artery **16** branches to the left vertebral artery **24**, usually in the shoulder area. The left subclavian artery **16** supplies blood to and may be directly accessed from (termed left radial access) the left arm.

(32) Four of the arteries illustrated in FIG. **1** supply blood to the cerebral vasculature: (1) the left carotid artery **14** (about 40% of cerebral blood supply); (2) the right carotid artery **18** (about 40% of cerebral blood supply); (3) the right vertebral artery **20** (about 10% of cerebral blood supply); and (4) the left vertebral artery **24** (about 10% of cerebral blood supply). The devices and methods described herein are also compatible with the prevalent (27%) bovine variant.

(33) FIG. **1** additionally illustrates an example three vessel protection system **100**, extending via right radial access across the top of the aortic arch and into the left subclavian artery **16**. While the illustrative protection system **100** is described as introduced via right radial access, it is contemplated that the protection system **100** may also be advanced via left radial access. In such an instance, the protection system **100** may be advanced from the left subclavian artery across the top of the aortic arch **10** and into the innominate artery **12**.

(34) Generally, the protection system **100** may include an outer sheath **110**, an inner member **118**, a guidewire **102**, a distal filter assembly **112** (see, for example, FIG. **2**), and a proximal filter assembly **124** (see, for example, FIG. **3**). The outer sheath **110** and the inner member **118** may be guided over the guidewire **102**, or along the guidewire **102** (in a rapid exchange configuration) to position a distal end of the outer sheath **110** within the left subclavian artery **16**. The inner member **118** may be radially inward of the outer sheath **110** and separately actuatable therefrom (e.g., slidably disposed within a lumen of the outer sheath **110**). It is contemplated that the guidewire **102** may be positioned within a lumen of the inner member **118** or within a lumen of the outer sheath **110**, as desired. In some cases, the lumen of the inner member **118** may be configured to receive a 0.014″ (0.3556 millimeter) diameter guidewire **102**, although other diameter guidewires **102** may be used, as desired. Each of the guidewire **102**, the outer sheath **110**, and the inner member **118** may be separately actuatable.

(35) The protection system **100** may include a distal end region **101** and a proximal end region **103**. The guidewire **102**, the outer sheath **110**, and the inner member **118** may be configured to extend

from the proximal end region **103** to the distal end region **101**. The proximal end region **103** may be configured to be held and manipulated by a user such as a surgeon. The distal end region **101** may be configured to be positioned at a target location such as, but not limited to, the left subclavian artery **16** (or the innominate artery **12** for a left radial access procedure). The proximal end region **103** may include a. **105**, a control **107** such as a slider, the outer sheath **110**, a port **109**, an inner member translation control **111**, such as a knob, and a hemostasis valve control **113** such as a knob. In some embodiments, the handle **105** may include fewer or more control elements than those illustrated in FIG. **1**. For example, the handle **105** may include controls such that the inner member **118** may be translated (e.g., distally advanced or proximally retracted) relative to the outer sheath **110** and/or the handle **105**, the outer sheath **110** can be translated relative to the handle **105** and/or the inner member **118**, the outer sheath **110** can be bent or deflected, etc. The handle **105** may include other controls for translating and/or deflecting other components of the system **100**, as will be described herein. The proximal end region **103** may also include the inner member **118** radially inward of the outer sheath **110**. While not explicitly shown, the proximal end region **103** may also include a filter wire (not explicitly shown in FIG. **1**) radially inward of the outer sheath **110** (and sometimes radially outward of the inner member **118**). Some illustrative filter wires are described in commonly assigned U.S. Pat. No. 9,566,144, the entirety of which is hereby incorporated by reference.
(36) The slider **107** can be used to translate the outer sheath **110** and/or a filter assembly (not explicitly shown in FIG. **1**). For example, the slider **107** may proximally retract the outer sheath **110**, the slider **107** may distally advance the filter assembly out of the outer sheath **110**, or the slider **107** may proximally retract the outer sheath **110** and distally advance the filter assembly (e.g., simultaneously or serially), which can allow the filter assembly to radially expand, which will be described in more detail herein. The slider **107** may also be configured to have an opposite translation effect, which can allow the filter assembly to be radially collapsed (e.g., due to compression by the outer sheath **110**) as the filter assembly is drawn into the outer sheath **110**. Other deployment systems are also possible, for example comprising gears or other features such as helical tracks (e.g., configured to compensate for any differential lengthening due to foreshortening of the filter assembly, configured to convert rotational motion into longitudinal motion), a mechanical element, a pneumatic element, a hydraulic element, etc. for opening and/or closing the filter assembly. The slider **107** may be independent of the inner member **118** such that the inner member **118** is longitudinally movable independent of the outer sheath **110** (and/or the filter wire). The inner member translation control **111** can be used to longitudinally translate the inner member **118**, for example before, after, and/or during deployment of the filter assembly. An inner member translation control **111** may comprise a slider in the handle **105** (e.g., separate from the slider **107**).
(37) A port **109** may be in fluid communication with the outer sheath **110** (e.g., via a Y-shaped connector in the handle **105**). The port **109** can be used to flush the device (e.g., with saline) before, during, and/or after use, for example to remove air. The port **109** can additionally, or alternatively, be used to monitor blood pressure at the target location, for example, by connecting an arterial pressure monitoring device in fluid communication with a lumen of the outer sheath **110**. The port **109** can be also or alternatively be used to inject contrast agent, dye, thrombolytic agents such as tissue plasminogen activator (t-PA), etc.
(38) A rotatable hemostasis valve control **113** can be used to reduce or minimize fluid loss through the protection system **100** during use. For example, a proximal portion and/or intermediate region of the protection system may be positioned in the right subclavian artery **22** and the direction of blood flow with respect to the system **100** will be distal to proximal, so blood may be otherwise inclined to follow the pressure drop out of the system **100**. The hemostasis valve control **113** is illustrated as being rotatable, but other arrangements are also possible (e.g., longitudinally displaceable). The hemostasis valve control **113** may be configured to fix relative positions of the outer sheath **110** and the filter assembly, for example as described with respect to the hemostasis

valve in U.S. Pat. No. 8,876,796. The hemostasis valve **113** may comprise, for example, an elastomeric seal and HV nut.

(39) To position the protection system **100** within the body, an incision may be made in the subject's right radial artery, or alternatively the right brachial artery. An introducer (not explicitly shown) may be placed into the incision and the protection system **100** inserted into the introducer. The protection system **100** may be advanced through the vasculature in a delivery configuration in which a filter assembly is collapsed or sheathed within the outer sheath **110**. The outer sheath **110**, inner member **118**, and the filter assemblies **112**, **124** may be advanced over a guidewire **102** to the target location. In some cases, the guidewire **102** may be positioned at the target location and the outer sheath **110**, inner member **118**, and the filter systems **112**, **124** subsequently advanced over the guidewire **102** to the target location. In other instances, the guidewire **102** and the outer sheath **110**, inner member **118**, and the filter systems **112**, **124** may be advanced substantially simultaneously with the guidewire **102** leading (e.g., positioned most distal to) the outer sheath **110**, inner member **118**, and the filter systems **112**, **124**.

(40) The delivery catheter or outer sheath **110** may have a diameter sized to navigate the vasculature from the incision to the target location. An outer sheath **110** sized for right radial access may have a diameter in the range of about 6 French (F). An outer sheath **110** sized for femoral access may be larger, although this is not required. These are just some examples. In some cases, the outer sheath **110** (or other components of the protection system **100**) may include deflectable tip or an articulatable distal tip. It is contemplated that the distal tip of the outer sheath **110** may be deflected (e.g., using the handle **105**) to facilitate navigation of the system **100** through the vasculature and cannulation of the target vessel. In some cases, the deflectable tip may be configured to deflect by about 90°. However, the tip of the outer sheath **110** may be made to deflect by less than 90 or more than 90, as desired. As described above, the deflection may be controlled by a rotating knob at the handle **105**, or the control mechanism. In some cases, the outer sheath **110** may include a pre-shaped and/or non-deflectable tip.

(41) The left subclavian artery **16** may be cannulated through a combination of advancing and torqueing the catheter **110**. In some cases, the left subclavian artery **16** may first be cannulated using the guidewire **102**. It is contemplated that in some cases, the deflection of the catheter tip may be relaxed to advance the outer sheath **110** into the left subclavian artery **16**. Once the protection system **100** (e.g., the distal end **104** of the outer sheath **110**) has been advanced to the target location, which in the illustrated embodiment may be the left subclavian artery **16**, the outer sheath **110** may be proximally retracted to expose a distal filter assembly **112** beyond the distal end **104** of the outer sheath **110**, as shown in FIG. **2**. Alternatively, or additionally, the inner member **118** may be distally advanced to deploy the distal filter assembly **112**. The distal filter assembly **112** may be configured to deployed in the distal-most great vessel relative to the location of the incision. For example, when right radial access is used, the distal-most great vessel is the left subclavian artery **16**. However, if left radial access is used, the distal-most great vessel is the innominate artery **12**.

(42) Referring additionally to FIG. **2**A, which illustrates an enlarged view of the distal filter assembly **112**, the distal filter assembly **112** may comprise a self-expanding filter assembly (e.g., comprising a superelastic material with stress-induced martensite due to confinement in the outer sheath **110**). The distal filter assembly **112** may comprise a shape-memory material configured to self-expand upon a temperature change (e.g., heating to body temperature). The distal filter assembly **112** may comprise a shape-memory or superelastic frame **114** (e.g., comprising a nitinol hoop) and a microporous filter element **120** (e.g., comprising a polymer including laser-drilled holes) coupled to the frame, for example similar to the filter assemblies described in U.S. Pat. No. 8,876,796.

(43) The frame **114** may generally provide expansion support to the filter element **120** in the expanded state. In the expanded state, the filter element **120** is configured to filter fluid (e.g.,

blood) flowing through the filter element **120** and to inhibit or prevent particles (e.g., embolic material) from flowing through the filter element **120** by capturing the particles in the filter element **120**. The frame **114** may configured to anchor the distal filter assembly **112** by engaging or apposing the inner walls of a lumen (e.g., blood vessel) in which the distal filter assembly **112** is expanded. In some cases, the distal filter assembly **112** may also anchor the proximal filter assembly **124**. For example. the distal filter assembly **112** may be a distal anchoring mechanism. The anchoring mechanism may include a filter membrane or may lack a filter membrane, as desired. As will be described in more detail herein, other anchoring mechanisms may be provided in addition to or alternatively to the distal filter assembly **112**. The frame **114** may comprise or be constructed of, for example, nickel titanium (e.g., nitinol), nickel titanium niobium, chromium cobalt (e.g., MP35N, 35NLT), copper aluminum nickel, iron manganese silicon, silver cadmium, gold cadmium, copper tin, copper zinc, copper zinc silicon, copper zinc aluminum, copper zinc tin, iron platinum, manganese copper, platinum alloys, cobalt nickel aluminum, cobalt nickel gallium, nickel iron gallium, titanium palladium, nickel manganese gallium, stainless steel, combinations thereof, and the like. The frame **114** may comprise a wire (e.g., having a round (e.g., circular, elliptical) or polygonal (e.g., square, rectangular) cross-section). For example, in some embodiments, the frame **114** may comprise a straight piece of nitinol wire shape set into a circular or oblong hoop or hoop with one or two straight legs running longitudinally along or at an angle to a longitudinal axis of the distal filter assembly **112**. The frame **114** may be coupled to a support strut **121**. If so provided, the straight legs may be on a long side of the distal filter assembly **112** and/or on a short side of the distal filter assembly **112**. The frame **114** may form a shape of an opening **115** of the distal filter assembly **112**. The opening **115** may be circular, elliptical, or any shape that can appropriately appose sidewalls of a vessel such as the left subclavian artery **16**. The distal filter assembly **112** may have a generally proximally-facing opening **115**. In other embodiments, the opening **115** may be distally facing. For example, the orientation of the opening **115** may vary depending on where the access incision is located.

(44) The frame **114** may include a radiopaque marker such as a small coil wrapped around or coupled to the hoop to aid in visualization under fluoroscopy. In some embodiments, the frame **114** may comprise a shape other than a hoop, for example, a spiral. In some embodiments, the distal filter assembly **112** may not include or be substantially free of a frame.

(45) In some embodiments, the frame **114** and the filter element **120** form an oblique truncated cone having a non-uniform or unequal length around and along the length of the distal filter assembly **112**. In such a configuration (e.g., along the lines of a windsock), the filter assembly **112** has a larger opening **115** (upstream) diameter and a reduced ending (downstream) diameter.

(46) The filter element **120** may include pores configured to allow blood to flow through the filter element **120**, but that are small enough to inhibit prevent particles such as embolic material from passing through the filter element **120**. The filter element **120** may comprise a filter membrane such as a polymer (e.g., polyurethane, polytetrafluoroethylene (PTFE)) film mounted to the frame **114**. The filter element **120** may have a thickness between about 0.0001 inches (0.00254 millimeters) and about 0.03 inches (0.76 millimeters) (e.g., no more than about 0.0001 inches (0.00254 millimeters), about 0.001 inches (0.0254 millimeters), about 0.005 inches (0. millimeters), about 0.01 inches (0.254 millimeters), about 0.015 inches (0.381 millimeters), about 0.02 inches (0.51 millimeters), about 0.025 inches (0.635 millimeters), about 0.03 inches (0.76 millimeters), ranges between such values, etc.).

(47) The film may comprise a plurality of pores or holes or apertures extending through the film. The film may be formed by weaving or braiding filaments or membranes and the pores may be spaces between the filaments or membranes. The filaments or membranes may comprise the same material or may include other materials (e.g., polymers, non-polymer materials such as metal, alloys such as nitinol, stainless steel, etc.). The pores of the filter element **120** are configured to allow fluid (e.g., blood) to pass through the filter element **120** and to resist the passage of embolic

material that is carried by the fluid. The pores can be circular, elliptical, square, triangular, or other geometric shapes. Certain shapes such as an equilateral triangular, squares, and slots may provide geometric advantage, for example restricting a part larger than an inscribed circle but providing an area for fluid flow nearly twice as large, making the shape more efficient in filtration verses fluid volume. The pores may be laser drilled into or through the filter element **120**, although other methods are also possible (e.g., piercing with microneedles, loose braiding or weaving). The pores may have a lateral dimension (e.g., diameter) between about 10 micron (μm) and about 1 mm (e.g., no more than about 10 μm, about 50 μm, about 100 μm, about μm, about 150 μm, about 200 μm, about 250 μm, about 300 μm, about 400 μm, about 500 μm, about 750 μm, about 1 mm, ranges between such values, etc.). Other pore sizes are also possible, for example depending on the desired minimum size of material to be captured.
(48) The material of the filter element **120** may comprise a smooth and/or textured surface that is folded or contracted into the delivery state by tension or compression into a lumen. A reinforcement fabric may be added to or embedded in the filter element **120** to accommodate stresses placed on the filter element **120** during compression. A reinforcement fabric may reduce the stretching that may occur during deployment and/or retraction of the distal filter assembly **112**. The embedded fabric may promote a folding of the filter to facilitate capture of embolic debris and enable recapture of an elastomeric membrane. The reinforcement material could comprise, for example, a polymer and/or metal weave to add localized strength. The reinforcement material could be imbedded into the filter element **120** to reduce thickness. For example, imbedded reinforcement material could comprise a polyester weave mounted to a portion of the filter element **120** near the longitudinal elements of the frame **114** where tensile forces act upon the frame **114** and filter element **120** during deployment and retraction of the distal filter assembly **112** from the outer sheath **110**.
(49) The distal filter assembly **112** may be coupled (e.g., crimped, welded, soldered, etc.) to a distal end of a deployment wire or filter wire (not explicitly shown) via a strut or wire **121**, although this is not required. When both or all of the filter wire and the strut **121** are provided, the filter wire and the strut **121** may be coupled to the inner member **118** proximal to the filter assembly **112** using a crimp mechanism **119**. In other embodiments, the filter wire and the strut **121** may be a single unitary structure. The filter wire and/or strut **121** can comprise a rectangular ribbon, a round (e.g., circular, elliptical) filament, a portion of a hypotube, a braided structure (e.g., as described herein), combinations thereof, and the like. A distal filter bag termination **116** or a distal end **116** of the distal filter assembly may also be coupled to the inner member **118**. In some cases, the distal end **116** may be coupled to the inner member using a nose cone **117**. The nose cone **117** may both secure the distal end **116** of the distal filter assembly **112** and reduce injury and/or vessel perforation during insertion of the system **100**. In some cases, the inner member **118** may reduce in diameter in the distal direction. The diameter may be step-wise or gradual.
(50) The distal filter assembly **112** in an expanded, unconstrained state may have a maximum diameter or effective diameter (e.g., if the mouth is in the shape of an ellipse the effective diameter is the diameter of the approximate circular opening **115** of the filter viewed from an end view) The diameter can be between about 1 mm and about 15 mm (e.g., at least about 2 mm, about 4 mm, about 6 mm, about 8 mm, about 10 mm or more, but generally less than about 15 mm or 12 mm or less depending upon the intended target vessel. In some embodiments (e.g., when the distal filter assembly **112** is configured to be positioned in the left subclavian artery **16**), the diameter may be between about 7 mm and about 12 mm (e.g., about 7 mm, about 8 mm, about 9 mm, about 10 mm, about 11 mm, about 12 mm, ranges between such values, etc.). In some embodiments (e.g., when the distal filter assembly **112** is configured to be positioned in the left vertebral artery **24**), the diameter may be between about 2 mm and about 4.5 mm (e.g., about 2 mm, about 2.5 mm, about 3 mm, about 3.5 mm, about 4 mm, about 4.5 mm, ranges between such values, etc.). Other diameters d or other types of lateral dimensions are also possible. Different diameters can allow treatment of

a selection of subjects having different vessel sizes.
(51) The filter assembly **112** may have a maximum length from a proximal limit of hoop **114** to the distal end or point of convergence **116** with the inner member **118**. The length can be between about 7 mm and about 50 mm (e.g., at least about 7 mm, about 8 mm, about 10 mm, about 12 mm, 16 mm, about 20 mm or more, but generally less than about 40 mm or 30 mm or 20 mm or less depending upon the intended target vessel. Other lengths are also possible, for example based on the diameter or effective diameter. For example, the length of the distal filter assembly **112** may increase as the diameter increases, and the length of the distal filter assembly **112** may decrease as the diameter decreases.
(52) In the illustrated embodiment, as described herein, the distal filter assembly **112** comprises a self-expanding support such as hoop or frame **114** which supports the open proximal end of an approximately conical filter membrane **120**. A support strut **121** may be provided to connect the hoop **114** to a portion of the system **100** such as the inner member **118**, to facilitate resheathing of the filter assembly **112** and also to orientate the hoop or frame **114**. In the illustrated embodiment, the distal filter assembly **112** may function both to filter blood entering the left subclavian artery **16**, and also to anchor the system as will be discussed in more detail herein.
(53) Continued proximal actuation of the outer sheath **110** while fixing the inner member **118** relative to the handle **105** (and thus the deployment wire and proximal bond, as will be described in more detail herein) may then deploy the aortic arch component or proximal filter assembly **124**, as shown in FIG. **3**. It is contemplated that distal end **104** of the outer sheath **110** may be withdrawn into the innominate artery **12** or the right subclavian artery **22** to fully expose the proximal filter assembly **124**. As the outer sheath **110** is withdrawn, the distal filter assembly **112** remains in the left subclavian artery **16**, supported by the inner member **118** and helping to anchor the proximal filter assembly **124**.
(54) The proximal filter assembly **124** may include an aortic ring **126** or support element and a filter membrane or filter element **128**. The aortic ring **126** may be similar in form and function to the frame **114** of the distal filter assembly **112**, although larger in scale. Similarly, the filter membrane **128** may be similar in form and function to the filter element **120** described in herein. The aortic ring **126** generally provides expansion support to the filter membrane **128** in its expanded configuration, while the filter membrane **128** is adapted to filter fluid, such as blood, and trap particles flowing therethrough. The aortic ring **126** may include an approximately elliptical hoop of a shape memory wire such as nitinol. The aortic ring **126** may be configured to engage the wall of the aorta and seal against the roof of the aortic arch outside of the ostia of all three great vessels **12**, **14**, **16** to isolate the cerebral vasculature. For example, the aortic ring **126** may be sized and shaped such that it extends from the ostium of the left subclavian artery **16** to the ostium of the innominate artery **12**. The center **125** of the aortic ring **126** is open to allow blood and debris to enter the device **124**, and the ring **126** is attached around its perimeter to the filter membrane **128**. The aortic ring **126** may be self-expanding such that as the outer sheath **110** is proximally withdrawn, the aortic ring **126** automatically expands, although this is not required. As will be described in more detail herein, the aortic ring **126** may be coupled to different portions of the protection system **100**.
(55) Once the proximal filter assembly **124** has been deployed, the inner member **118** may be retracted to minimize intrusion into the aorta **10**, as illustrated in FIG. **4**. The position of the inner member **118** may be adjusted using the control **111** at the handle **105**. As will be described in more detail herein, the mechanism coupling the proximal filter **124** to the system **100** may be adjusted by proximally retracting or distally advancing the handle **105** to ensure the proximal filter **124** is fully deployed, the aortic ring **126** is apposed to the roof of the aortic arch **10**, and the inner member **118** is pulled up against the roof of the aortic arch **10** to ensure that the membrane **128** is fully against the roof of the arch **10**. The proximal filter assembly **124** is thus positioned to protect the cerebral vasculature **14**, **18**, **20**, **24** from embolic debris during an endovascular procedure such as TAVR.

(56) Once the protection system **100** has been deployed, the TAVR, or other index procedure, may be performed. Once the index procedure is contemplate, the inner member **118** may be advanced relative to the handle **105** to apply tension to the proximal filter element **128** and/or the distal filter element **120**. The outer sheath **110** may then be advanced relative to the handle **105**, while fixing the position of the inner member **118** relative to the handle **105**, until the proximal filter assembly **124** and the distal filter assembly **112** are fully sheathed. The protection system **100** may then be removed from the introducer sheath.
(57) FIGS. **5**A and **5**B illustrate a first mechanism for coupling the proximal filter assembly **124** to the protection system. In the illustrated embodiment, a proximal end **130** of the aortic ring **126** is connected to a proximal bond tube **132** with an aortic ring connector **134**. The proximal bond tube **132** may be a tubular structure configured to link or couple several structures of the system **100**. In some cases, the structures coupled to the proximal bond tube **132** may extend into and be coupled within a lumen of the proximal bond tube **132**, as will be described in more detail herein. The aortic ring connector **134** may be formed as a monolithic or unitary structure with the aortic ring **126**. Alternatively, the aortic ring connector **134** may be separately formed and coupled with the aortic ring **126**. It is contemplated that the aortic ring connector **134** may be formed from a nitinol wire, or other suitable material. The proximal end region **136** of the membrane **128**, or the proximal filter bag termination **136**, may also be connected to the proximal bond tube **132**.
(58) The proximal bond tube **132** may be connected to a delivery wire **138** configured to extend proximally to the handle **105**. The handle **105** may include at least three controls (which may be sliding controls) including a control for translation of the inner member **118** relative to the handle **105**, a control for translation of the delivery catheter **110** relative to the handle **105** to allow for deployment and retrieval of the filter assemblies **112**, **124**, and a control for deflection of the distal tip of the delivery catheter **110**. The delivery wire **138** is connected to both the proximal bond tube **132** and the handle **105**. Thus the aortic ring **126** and proximal filter bag termination **136** positions are fixed relative to the handle **105**. As such, the position of the proximal bond tube **132** relative to the patient's body can be moved by either advancing or retracting the handle **105**. As described above, the distal filter frame **114**, and thus the distal end region **116** of the distal filter element **120** or the distal filter bag termination **116**, are connected to the inner member **118**, and are advanced and retracted relative to the outer sheath **110** with a first handle control **111**. The steerable outer sheath or delivery catheter **110** may be withdrawn relative to the handle **105** to deploy the device using a second handle control **107**. The tip **104** of the outer sheath or delivery catheter **110** may be deflected to assist with cannulation of the left subclavian artery **16** (or the brachiocephalic artery **12**) using a third handle control (not explicitly shown).
(59) In FIG. **5**A, the aortic ring connector **134** may be formed by bonding two proximally extending sections of the wire forming the hoop of aortic ring **126** to produce a single, relatively more rigid strut **134**. Alternatively, as shown in FIG. **5**B, the two wires **134**, **135** which extend proximally from aortic ring **126** may be movable with respect to each other. This allows the minor axis of the aortic ring **126** to self adjust to the aorta **10**, as the two wires **134**, **135** move towards or away from each other. Either an excess of membrane **128**, or overlapping sections of membrane **128** carried by the wires **134**, **135** allow continuity of the membrane **128** coverage regardless of the spacing between wires **134** and **135**.
(60) FIG. **6** is a partial cross-sectional view of the illustrative system **100** taken at line **6-6** of FIG. **5**A. As can be seen, the proximal bond tube **132** defines a lumen **133**. The inner member **118** may be sliably disposed within the lumen **133** such that the inner member **118** is separately movable from the proximal bond tube **132**. The proximal filter bag termination **136** may be coupled to an inner surface of the proximal bond tube **132**. However, other coupling configurations can be used, as desired. Further, the deployment or control wire **138** may also be coupled to the proximal bond. The control wire **138** may be coupled directly to the proximal bond or indirectly coupled (e.g., with the proximal filter bag termination **136** disposed therebetween), as desired. In some cases, the

control wire **138** and the aortic ring connector **134** may form a unitary structure. In other embodiments, the deployment or control wire **138** and the aortic ring connector **134** may be separate and distinct components.

(61) FIG. **7** is a cross-sectional view of the protection system **100** taken at line **7-7** of FIG. **5**A. As can be seen, the membrane **128** may be coupled to the aortic ring **126** about its perimeter to define an opening for **125** that allows blood and debris to enter the proximal filter assembly **124**. The membrane **128** also forms a net that prevents debris from exiting the proximal filter assembly **124**. In some cases, the aortic ring **126** may be include a radiopaque coil or marker to facilitate to allow the aortic ring **126** to be visualized.

(62) As described herein, a distal steering segment of steerable delivery catheter **110** may be deflected laterally in response to manipulation of a steering control on the proximal manifold or handle **105**. Referring additionally to FIG. **8**, which illustrates a cross-sectional view of the illustrative system **100**, taken at line **8-8** of FIG. **5**A, in the illustrated embodiment, the sidewall of delivery catheter or outer sheath **110** is provided with a pull wire lumen **144** for axially moveably receiving a pull wire **140** which may be proximally retracted to deflect the distal tip laterally. A second pull wire may alternatively be provided, such as at 180 degrees around the catheter **110** from the first pull wire, to facilitate straightening or deflecting the catheter in a second, opposite direction. Further, in some embodiments, the outer sheath **110** may include a reinforcement braid **131**, although this is not required.

(63) FIG. **9** illustrates another alternative embodiment for the system of FIGS. **1-8**. In the system **100′**, the handle **105** may include at least three controls (which may be sliding controls) including a control for translation of the inner member **118** relative to the handle **105**, a control for translation of the delivery catheter **110** relative to the handle **105** to allow for deployment and retrieval of the filter assemblies **112**, **124**, and a control for deflection of the distal tip of the delivery catheter **110**. In the system **100′**, the proximal filter bag termination **136** is connected to the proximal bond tube **132**. The delivery wire **138** is also connected to the proximal bond tube **132** and the handle **105**, as can be seen in FIG. **11** which is a partial cross-sectional view of the system **100′** taken at line **11-11** of FIG. **9**. Thus, movement of the handle **105** is translated to the proximal bond tube **132**. However, in the embodiment of FIG. **9**, the proximal end **130** of the aortic ring **126** is connected to the inner member **118** via the aortic ring connectors **134**, **135**. For example, referring to FIG. **10**, which is a partial cross-sectional view of the system **100′** taken at line **10-10** of FIG. **9**, the aortic ring connectors **134**, **135** may be coupled to the inner member **118** via a connection tube **137**, which may be a heat shrink tube. The distal end **139** of the aortic ring **126** may be coupled to the distal filter frame **114** via a distal support strut **142**. Thus, the aortic ring **126**, the distal filter frame **114**, the distal filter membrane **120**, and the inner member **118** may all be advanced and retracted relative to the delivery catheter or outer sheath **110** with the first control element **111** in the handle **105**. The steerable delivery catheter **110** may be withdrawn relative to the handle **105** to deploy the devices **112**, **124** using a second handle control **107**. The tip of the delivery catheter **110** may be deflected to assist with cannulation of the left subclavian artery **16** (or the brachiocephalic artery **12**) using a third handle control (not explicitly shown).

(64) FIG. **12** illustrates another alternative embodiment for the system of FIGS. **1-8**. In the illustrative system **100″** of FIG. **12**, the handle **105** may include at least three controls (which may be sliding controls) including a control for translation of the inner member **118** relative to the handle **105**, a control for translation of the delivery catheter **110** relative to the handle **105** to allow for deployment and retrieval of the filter assemblies **112**, **124**, and a control for deflection of the distal tip of the delivery catheter **110**. The proximal filter bag termination **136** is connected to the proximal bond tube **132**. The delivery wire **138** and the aortic ring connector **134** are also connected to the proximal bond tube **132** and the handle **105**. Thus, movement of the handle **105** is translated to the proximal bond tube **132**. In the embodiment of FIG. **12**, the distal end **139** of the aortic ring **126** may be coupled to the distal filter frame **114** via a distal support strut **142**. Thus, the

proximal filter bag termination **136**, the aortic ring **126** and the distal filter frame **114** are all fixed relative to the handle **105**. In other words, movement of the handle **105** is translated to the proximal bond tube **132** via the delivery wire **138** and as the aortic ring **126** is coupled to the proximal bond tube **132** and the distal filter frame, the aortic ring **126** and the distal filter frame **114** also move with the handle **105**. The distal filter bag termination **116** is connected to the inner member **118** and is advanced and retracted relative to the delivery catheter **110** with a first handle control **111** (e.g., via movement of the inner member **118**). The steerable delivery catheter **110** may be withdrawn relative to the handle **105** to deploy the device using a second handle control **107**. The tip of the delivery catheter **110** may be deflected to assist with cannulation of the left subclavian artery **16** (or the brachiocephalic artery **12**) using a third handle control (not explicitly shown).

(65) FIG. **13** illustrates another alternative embodiment for the system of FIGS. **1-8**. In the illustrative system **100′′′** of FIG. **13**, the proximal filter bag termination **136** is connected to a first proximal bond tube **132** and is fixed relative to the handle **105** via the deployment wire **138** which is also coupled to the first proximal bond tube **132** and the handle **105**. The distal filter frame **114**, and thus the distal filter bag termination **116**, are connected to the inner member **118**. Both the distal filter frame **114** and the distal filter bag termination **116** are advanced and retracted relative to the delivery catheter **110** with a first handle control **111**. The steerable delivery catheter **110** may be withdrawn relative to the handle **105** to deploy the device **112**, **124** using a second handle control **107**. The tip of the delivery catheter **110** may be deflected to assist with cannulation of the left subclavian artery **16** (or the brachiocephalic artery **12**) using a third handle control (not explicitly shown). The proximal end **130** of the aortic ring **126** is connected to a sliding aortic ring bond tube **143** via the aortic ring connector **134**. The second sliding bond tube **143** is also coupled to an aortic ring deployment wire **145**. The second sliding bond tube **143** may be advanced and retracted relative to the delivery catheter **110** with a fourth handle control (not explicitly shown).

(66) The handle **105** of the system **100′′′** may include at least four controls (which may be sliding controls) including a control for translation of the inner member **118** relative to the handle **105**, a control for translation of the delivery catheter **110** relative to the handle **105** to allow for deployment and retrieval of the filter assemblies **112**, **124**, a control for deflection of the distal tip of the delivery catheter **110**, and a control for translation of the aortic ring deployment wire **145** relative to handle **105**. This may allow the aortic ring connector **134** (and thus the aortic ring **126**), the inner member **118** and the proximal bond tube **132** (and thus the proximal filter membrane **128**) to all be adjusted relative to each other. This may allow the apposition of the aortic ring **126** to the roof of the aortic arch **10** to be further optimized. In one technique, the position of the aortic ring connector **134** may be fixed, and the position of the proximal bond tube **132** (and thus the proximal termination of the filter membrane **136**), to be withdrawn towards the handle **105**. This applied tension on the proximal membrane **128** may draw the aortic ring **126** towards the roof or the aortic arch **10**, and increase the bend in the aortic ring connector **134**, thus improving apposition between the aortic ring **126** and the roof of the aortic arch **10**.

(67) FIG. **14** illustrates a cross-sectional view of the illustrative system **100′′′**, taken at line **14-14** of FIG. **13**. In the illustrated embodiment, the sidewall of delivery catheter or outer sheath **110** is provided with a pull wire lumen **144** for axially moveably receiving a pull wire **140** which may be proximally retracted to deflect the distal tip laterally. A second pull wire may alternatively be provided, such as at 180 degrees around the catheter **110** from the first pull wire, to facilitate straightening or deflecting the catheter in a second, opposite direction. Further, in some embodiments, the outer sheath **110** may include a reinforcement braid **131**, although this is not required. The deployment wire **138** and the aortic ring deployment wire **145** may extend within the lumen of the outer sheath **110** but exterior to the inner member **118**.

(68) FIG. **15** is a cross-sectional view of the illustrative system **100′′′** taken at line **15-15** of FIG. **13**. As can be seen, the proximal bond tube **132** defines a lumen **133**. The inner member **118** may be slidably disposed within the lumen **133** such that the inner member **118** is separately movable

from the proximal bond tube **132**. Further, the aortic ring deployment wire **145** may also extend through the lumen **133** such that the aortic ring deployment wire **145** (and thus the second bond tube **143**) is separately movable from the proximal bond tube **132**. The proximal filter bag termination **136** may be coupled to an inner surface of the proximal body **132**. However, other coupling configurations can be used, as desired. Further, the deployment or control wire **138** may also be coupled to the proximal bond tube **132**. The control wire **138** may be coupled directly to the proximal bond or indirectly coupled (e.g., with the proximal filter bag termination **136** disposed therebetween), as desired. Both the proximal bond tube **132** and the aortic ring bond tube **143** may slide freely over (and/or relative to) the inner member **118**.
(69) In the protection systems **100**, **100′**, **100″**, **100‴** it may be desirable for correct deployment, functioning and retrieval of the filter assemblies **112**, **124** for the rotational orientation between the delivery catheter **110**, the proximal bond tube **132**, and the inner member **118** to be controlled. Control of the inner member **118** (and thus the distal filter assembly **112**) orientation relative to the aortic ring **126** (and thus the proximal bond tube **132**) may prevent the distal filter frame **114** from rotating relative to the aortic ring **126** and/or proximal bond tube **143132**, and resulting in a twist in the membrane **128** between the aortic ring **126** and the distal filter assembly **112**, leading to possible restriction of flow of blood and/or debris into the distal filter assembly **112** and thus the left subclavian artery. It may be desirable to control the inner member **118** (and thus the distal filter assembly **112**) and the aortic ring **126** (and thus proximal bond **132**) orientations relative to the delivery catheter **110**, and thus to the deflection direction of the deflectable outer sheath **110**, to ensure that the device **112**, **124** deploys from the tip of the delivery catheter **110** in the aortic arch **10** in the correct orientation.
(70) It is contemplated that the orientation may be controlled by fixing the rotational orientation of the deployment wire **138** (and thus the proximal bond tube **132**) relative to the inner member **118** at the attachment points within the handle **105**. If the orientation of the deployment wire **138** and the inner member **118**, where they are attached to the handle **105**, are fixed, the orientation of the proximal bond tube **132** and the distal filter assembly **112** may be controlled. However, the handle **105** may be separated from the distal filter assembly **112** by some distance, and the connecting inner member **118** and deployment wire **138** may twist or torque relative to each other resulting in a misalignment of the distal filter assembly **112** to the aortic ring **126**. This twisting may be reduced by replacing the deployment wire **138** with a deployment tube **150** that runs in the annular space between the delivery catheter **110** and the inner member **118**. FIG. **16** illustrates a cross-sectional view of this feature. Constructing the deployment wire from a tube **150** may have the advantage that a tube resists torque more effectively than a wire, and may result in better alignment between the proximal bond tube **132** and the aortic ring **126** as the tube structure **150** may rotationally deflect to a lesser degree than the wire. The tube **150** may be constructed of polymer, braid reinforced polymer, stainless steel or other metal. FIG. **17** illustrates a partial side view of an illustrative delivery tube **150** constructed of stainless steel, nitinol or other metal hypo tube with laser cut slots **152** to improve flexibility while maintaining axial and torsional stiffness to maintain the position (axially and rotationally) of the proximal bond tube **132** relative to the handle **105**.
(71) In another illustrative embodiment, the orientation may be controlled by keying the proximal bond tube **132** to the inner member **118** such that the proximal bond tube **132** can slide freely over the inner member **118**, but cannot rotate relative to the inner member **118**. For example, orientation of the proximal bond tube **132** relative to the inner member **118** may be controlled at the proximal bond tube **132** instead of, or in addition to, at the handle **105**. This may be done by creating a non-round or non-circular profile (e.g., a non-circular cross-sectional shape) of the inner member **118** in at least the region where the proximal bond tube **132** slides over the inner member **118**. This profile (e.g., cross-sectional shape) could be square, hexagonal, triangular, or other non-round profiles. The inner opening of the proximal bond tube **132** may be profiled or keyed to match the cross-section of the inner member **118** such that it can slide along the inner member **118**, but not rotate

relative to the inner member **118**. FIG. **18** illustrates a cross-sectional view of such a keyed arrangement between the inner member **118** and at least a portion of the proximal bond tube **132**. While not explicitly shown, when so provided, the aortic ring bond tube **143** and the inner member **118** may be similarly keyed. In some cases, the proximal bond tube **132** may have a circular outer profile while the shape of the lumen is keyed or shaped to match a shape of an outer surface of the inner member **118** to limit rotation of the inner member **118**.

(72) The inner member **118** may be constructed of a polymer, a metal, a composite, a wire, a braid reinforced polymer such as polyimide, nylon, Pebax, etc., or other suitable material. The inner member **118** may be constructed with variable stiffness so that, for example, the distal portion is less stiff than the proximal portion. However, this is not required. In some embodiments, limit stops may be added to the inner member **118** to limit travel of the aortic ring connector **134**, **135** (and thus the aortic ring **126**) relative to the inner member **118**, in both the proximal and distal directions. The inner member **118** may moves freely through the proximal bond tube **132**. The limit stops may prevent the operator from extending the inner member **118** to the extent that the inner member **118** would protrude out of the aortic ring **126** and/or prevent the operator from withdrawing the inner member **118** to the extent that the distal filter assembly **112** is pulled out of correct placement in the left subclavian artery **16**. By placing hard stops (e.g., structural features extending from an outer surface of the inner member **118**) on the inner member **118** on either side of the proximal bond tube **132**, the distance that an operator can pull or push on the inner member **118** can be controlled.

(73) The proximal end **136** of the proximal filter membrane **128** may be terminated at the proximal bond tube **132**. Apposition and sealing of the proximal filter membrane **128** in the brachiocephalic artery **12** may be achieved through a combination of sealing of the aortic ring **126** to the roof of the aortic arch **10**, and the expansion of the filter membrane **128** against the walls of the brachiocephalic artery **12** due to blood flow through the filter membrane **128**. In alternate embodiments, a proximal filter frame (e.g., similar in form and function to the distal filter frame **114**) may be added to ensure that the filter membrane **128** opens fully in the brachiocephalic artery **12**. This proximal filter frame may be constructed to provide positive circumferential contact between the membrane **128** and the walls of the brachiocephalic artery **12** or may function to ensure that the filter is open yet not circumferentially press the membrane **128** against the vessel walls. As with the distal filter frame **114**, the proximal filter frame may be bonded to the membrane or may freely float inside the membrane.

(74) In some embodiments, the self-expanding distal filter frame **114** may alternately not be bonded to the filter membrane **120** so that the distal filter frame **114** may rotate relative to the membrane **120** to prevent twisting of the membrane but would still act to hold the membrane **120** open inside the left subclavian artery **16** (or the brachiocephalic artery **12**). The inner member **118** may be extended past distal filter assembly **112** to stabilize distal filter assembly **112**. Also, the guidewire **102** may be extended beyond the distal filter assembly **112** to stabilize the distal filter assembly **112** in the left subclavian artery **16** and help maintain apposition of the filter frame to the vessel wall.

(75) The distal filter assembly **112** is anchored in the left subclavian artery **16** by the expansion of the distal filter frame **114** to stabilize the device during delivery of the proximal filter assembly **124**. Alternatively, or additionally, this anchoring function could be performed by an expanding anchor element that is inside of or distal to the filter membrane **120**. This could take the form of a self-expanding stent structure, expanding coil, inflatable balloon, etc. This feature may be helpful holding the position of the filter assembly **112** and thus the distal portion of the device when the inner member **118** is tensioned.

(76) FIG. **19** is a first illustrative alternative distal anchoring structure **160** positioned distal to the distal filter assembly **112**. A self-expanding nitinol mesh anchor **160** or similar component to help hold the position of the distal end of the device. This nitinol mesh **160** can be located between the distal filter assembly **112** and the tip or at the very tip of the catheter. The mesh anchor structure

160 can also be of any shape including but not limited to conical, cylindrical, etc.

(77) FIG. **20** illustrates another alternative distal anchoring structure including an expandable balloon **170** positioned distal to the distal filter assembly **112**. In some cases, the expandable balloon **170** may be located on one side of the inner member **118** distal to the distal filter assembly **112**. When in position, the balloon **170** can be expanded to help hold the position of the distal end of the inner member **118**. Placing the balloon **170** on one side of the inner member **118** may allow blood flow to continue along the other sides of the inner member **118**.

(78) FIG. **21** illustrates another alternative distal anchoring structure including one or more nitinol rings **180** that can expand after the outer sheath **110** is withdrawn. These nitinol rings **180** can then help anchor the distal end of the inner member **118** in the left subclavian artery **16**.

(79) The filter membranes **120**, **128** may be constructed of laser drilled or perforated polymer membrane, woven nitinol wire mesh, other metal mesh, woven PEEK or other polymer mesh, etc. The porosity of the filter membranes **120**, **128** may be around 140 microns, however other porosities, either larger or smaller, are possible. The membrane hole pattern may be uniform, or the membrane may be perforated selectively in given regions of the membrane **120**, **128** to either allow or restrict flow in those select regions to improve flow through lumens, encourage apposition, etc. The filter membrane **120**, **128** may have fabric or other reinforcing or stiffening in selected areas to improve sheathing, unsheathing, folding of the membrane and apposition performance, see For example, the proximal filter membrane **128** may include is a strip of fabric reinforcement in the membrane material that extends from the point where the aortic ring connector **134** meets the aortic ring **126** to the proximal bond tube **132**. The fabric reinforcement may prevent or reduce bunching of the membrane **128** during resheathing, and may help the membrane **128** fold in a controlled manner during sheathing.

(80) The aortic ring **126** and/or the distal filter frame **114** may be constructed of nitinol wire, as described herein, however they may constructed with other flexible materials such as polymer, stainless steel, a composite of multiple material, woven or braided like a cable, etc. The aortic ring **126** and/or the distal filter frame **114** may also be laser cut from a flat sheet of metal, polymer, etc. and then formed or shape set to the desired final shape.

(81) The aortic ring **126** may be constructed such that it is stiffer or more flexible in some section than in other. For example, if the aortic ring **126** is constructed of nitinol wire, it may be selectively ground to reduce the diameter or thickness to alter stiffness to improve performance, improve apposition, make sheathing easier, etc.

(82) Radiopaque markers may be added to the filter frame(s) **114**, the aortic ring **126**, the aortic ring connectors **134**, **135** and/or other portions of the device to allow visualization under fluoroscopy. The marker may be placed in selective locations or may be in the form of a coil placed over the entire aortic ring **126**, filter hoop **114** or connector structures **134**, **135** as needed to help with positioning and visualization.

(83) The aortic connector **134**, **135** may be constructed of more rigid materials such as nitinol, stainless steel, polymer, etc., or may be constructed of more flexible materials such as suture, fabric, or other flexible material that will still support tension to allow sheathing of the aortic ring but would allow improved apposition of aortic ring to the roof of the aorta. The aortic ring **126** may be constructed from a loop of material, for example nitinol wire, and thus the aortic ring connector **134**, **135** may be formed by the two ends of the loop, and then terminated at the proximal bond tube **132**. The mechanical characteristics of the aortic ring connector **134**, **135** may be altered by many methods including the addition of heat shrink **190** over the wires **134**, **135**, as illustrated in FIG. **22**A, the addition of a stiffening ribbon **192** (with or without a heat shrink layer **190**), as shown illustrated in FIG. **22**B, or using multiple layers of polymer. **190**, **194** (with or without a stiffening ribbon **192**), as illustrated in FIG. **22**C

(84) The aortic ring connector(s) **134**, **135** may be curved or shaped to improve aortic ring apposition on the proximal edge of the brachiocephalic **12** or left subclavian **16** artery ostia. The

aortic ring connector may be formed in a curved shape (as opposed to a straight connector) may allow the aortic ring **126** to fully cover the brachiocephalic artery **12** without leaving a gap on the proximal edge **130**.
(85) In some embodiments, the aortic ring connector **134**, **135** may improve apposition of the aortic ring **126** to the roof of the aortic arch **10** by for including a back-bend shape FIG. **23** illustrates an aortic ring connector **134** including curved shape that bends back on itself. In addition, the shape and bend of the aortic ring connector **134** may be adjustable. For example, the aortic ring connector **134** may include an adjustable tension member to allow the apposition of the aortic ring **126** to the roof of the aortic arch to be adjusting and optimized following deployment.
(86) While the methods and devices described herein may be susceptible to various modifications and alternative forms, specific examples thereof have been shown in the drawings and are described in detail herein. It should be understood, however, that the inventive subject matter is not to be limited to the particular forms or methods disclosed, but, to the contrary, covers all modifications, equivalents, and alternatives falling within the spirit and scope of the various implementations described and the appended claims. Further, the disclosure herein of any particular feature, aspect, method, property, characteristic, quality, attribute, element, or the like in connection with an implementation or embodiment can be used in all other implementations or embodiments set forth herein. In any methods disclosed herein, the acts or operations can be performed in any suitable sequence and are not necessarily limited to any particular disclosed sequence and not be performed in the order recited. Various operations can be described as multiple discrete operations in turn, in a manner that can be helpful in understanding certain embodiments; however, the order of description should not be construed to imply that these operations are order dependent. Additionally, the structures described herein can be embodied as integrated components or as separate components. For purposes of comparing various embodiments, certain aspects and advantages of these embodiments are described. Not necessarily all such aspects or advantages are achieved by any particular embodiment. Thus, for example, embodiments can be carried out in a manner that achieves or optimizes one advantage or group of advantages without necessarily achieving other advantages or groups of advantages. The methods disclosed herein may include certain actions taken by a practitioner; however, the methods can also include any third-party instruction of those actions, either expressly or by implication. For example, actions such as “deploying a self-expanding filter” include “instructing deployment of a self-expanding filter.” The ranges disclosed herein also encompass any and all overlap, sub-ranges, and combinations thereof. Language such as “up to,” “at least,” “greater than,” “less than,” “between,” and the like includes the number recited. Numbers preceded by a term such as “about” or “approximately” include the recited numbers and should be interpreted based on the circumstances (e.g., as accurate as reasonably possible under the circumstances, for example ±5%, ±10%, ±15%, etc.). For example, “about 7 mm” includes “7 mm.” Phrases preceded by a term such as “substantially” include the recited phrase and should be interpreted based on the circumstances (e.g., as much as reasonably possible under the circumstances). For example, “substantially straight” includes “straight.”
(87) Those skilled in the art will recognize that the present invention may be manifested in a variety of forms other than the specific embodiments described and contemplated herein. Accordingly, departure in form and detail may be made without departing from the scope and spirit of the present invention as described in the appended claims.

Claims

1. An embolic protection system for isolating a cerebral vasculature, the system comprising: an outer sheath, having a proximal end and a distal end; an inner member extending through and slidable within a lumen of the outer sheath; a distal anchoring mechanism fixed to a distal end region of the inner member; a proximal filter membrane carried between the inner member and the

outer sheath, the proximal filter membrane extending proximally from the distal anchoring mechanism and configured to extend from a left subclavian artery to a brachiocephalic artery; an aortic ring coupled to the proximal filter membrane and configured to seal the proximal filter membrane against a wall of an aorta; and a handle including a first actuation mechanism coupled to a proximal end of the inner member, wherein the first actuation mechanism is configured to retract the inner member proximally relative to the proximal filter membrane, pulling the inner member up against a roof of an aortic arch.
2. The embolic protection system of claim 1, wherein a center of the aortic ring is open to allow debris to enter the proximal filter membrane.
3. The embolic protection system of claim 1, wherein the distal anchoring mechanism includes a self-expanding frame.
4. The embolic protection system of claim 3, further comprising a distal filter membrane coupled to the self-expanding frame.
5. The embolic protection system of claim 4, wherein the distal anchoring mechanism includes an expandable anchor disposed distal of the distal filter membrane.
6. The embolic protection system of claim 5, wherein the expandable anchor includes a self-expandable nitinol mesh.
7. The embolic protection system of claim 5, wherein the expandable anchor includes a balloon.
8. The embolic protection system of claim 5, wherein the expandable anchor includes one or more self-expandable rings.
9. The embolic protection system of claim 1, further comprising a proximal bond tube coupled to a proximal end of the proximal filter membrane.
10. The embolic protection system of claim 9, wherein the inner member is slidably disposed through a lumen of the proximal bond tube.
11. The embolic protection system of claim 9, further comprising a connector coupling the aortic ring with the proximal bond tube.
12. The embolic protection system of claim 11, wherein the connector and the aortic ring are a single monolithic structure.
13. The embolic protection system of claim 11, wherein the proximal bond tube is coupled to a delivery wire extending proximally to the handle.
14. The embolic protection system of claim 10, wherein a cross-sectional shape of the inner member adjacent to the proximal bond tube is non-circular and at least a portion of the proximal bond tube has a non-circular cross-sectional shape such that the proximal bond tube cannot rotate relative to the inner member.
15. An embolic protection system for isolating a cerebral vasculature, the system comprising: an outer sheath, having a proximal end and a distal end; an inner member extending through and slidable within a lumen of the outer sheath; a self-expanding distal filter coupled fixed to a distal end region of the inner member; a proximal filter carried between the inner member and the outer sheath, the proximal filter extending proximally from the self-expanding distal filter and configured to extend from a left subclavian artery to a brachiocephalic artery and self-expanding to seal against a wall of an aorta, wherein the proximal filter has a central opening along one side configured to allow debris to enter the proximal filter from the aorta; and a handle including a first actuation mechanism coupled to a proximal end of the inner member, wherein the first actuation mechanism is configured to retract the inner member proximally relative to the proximal filter, pulling the inner member up against a roof of an aortic arch.
16. The embolic protection system of claim 15, further comprising a proximal bond tube coupled to a proximal end of the proximal filter.
17. The embolic protection system of claim 16, wherein the proximal bond tube is configured such that the inner member is slidably disposed through a lumen of the proximal bond tube and the proximal bond tube cannot rotate relative to the inner member.

18. The embolic protection system of claim 16, wherein the proximal bond tube is coupled to a delivery wire extending proximally to the handle.
19. The embolic protection system of claim 16, wherein the proximal filter includes an aortic ring and a proximal filter membrane, the aortic ring configured to seal the proximal filter membrane against the wall of the aorta, wherein the aortic ring is coupled to the proximal bond tube.
20. An embolic protection system for isolating a cerebral vasculature, the system comprising: an outer sheath, having a proximal end and a distal end; an inner member extending through and slidable within a lumen of the outer sheath; a distal anchor fixed to a distal end region of the inner member; a proximal filter membrane carried between the inner member and the outer sheath, the proximal filter membrane extending proximally from the distal anchor and configured to extend from a left subclavian artery to a brachiocephalic artery; an aortic ring coupled to the proximal filter membrane and configured to seal the proximal filter membrane against a wall of an aorta; and wherein the inner member is configured to be retractable proximally relative to the proximal filter membrane to pull the inner member up against a roof of an aortic arch.